9. Absorption Laws

(i) $p \vee (p \wedge q) \equiv p$ (ii) $p \wedge (p \vee q) \equiv p$

p	q	$p \wedge q$	$p \vee q$	$p \vee (p \wedge q)$	$p \wedge (p \vee q)$
T	T	T	T	T	T
T	F	F	T	T	T
F	T	F	T	F	F
F	F	F	F	F	F

Table 12.22

(i) The entries in the columns corresponding to $p \vee (p \wedge q)$ and p are identical and hence they are equivalent.

(ii) The entries in the columns corresponding to $p \wedge (p \vee q)$ and p are identical and hence they are equivalent.

Example 12.17

Establish the equivalence property: $p \rightarrow q \equiv \neg p \vee q$

Solution

p	q	$\neg p$	$p \rightarrow q$	$\neg p \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Table 12.23

The entries in the columns corresponding to $p \rightarrow q$ and $\neg p \vee q$ are identical and hence they are equivalent.

Example 12.18

Establish the equivalence property connecting the bi-conditional with conditional:

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

Solution

p	q	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

Table 12.24

The entries in the columns corresponding to $p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ are identical and hence they are equivalent.

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